4 The α-Relaxation Dynamics of Glycerol

Glycerol is perhaps one of the most widely studied glass forming systems. As a hydrogen bonding liquid, it exhibits more complicated behavior than a simple van der Waals liquid such as cumene discussed in Ch. 5. A topic of significant interest within the literature over the past twenty five years has been the behavior of fragility as a function of pressure in such hydrogen bonding systems. Whereas most molecular liquid systems exhibit either a decreasing fragility as pressure increases or very little pressure dependence at all, glycerol and other hydrogen bonding liquids have been observed to show an increase in fragility which has been attributed to the suppression of hydrogen bonding under pressure [64, 65, 66]. This chapter will explore the current state of the literature on this topic, apply a new model to the dynamics of glycerol over a wide range of pressure and temperature with the aid of literature data obtained through multiple methods, reassess the efficacy of viscosity data obtained by Cook *et al*. that has come under some scrutiny, further compare resulting thermodynamic scaling regimens with previous literature results, as well as provide a more precise measure of the pressure dependence of fragility in glycerol.

4.1 Earlier Works Regarding Glycerol Dynamics

As previously asserted, glycerol is one of the most widely studied glass forming systems, with the temperature dependence of its viscosity being investigated as far back as 1926 in the work of Tammann and Hesse [52]. However the modern history of glycerol, as relevant to this work, arguably begins with Davidson and Cole's investigation of the temperature dependence of the dielectric relaxation in glycerol circa 1951 [49]. Davidson and Cole found that not all associated

liquids can be modeled with a simple single-relaxation or Debye equation as shown in Eq. 4.1 [67].

$$\epsilon^* = \epsilon_\infty + \frac{\epsilon_0 - \epsilon_\infty}{1 + i\omega\tau_0}, \tag{4.1}$$

where ϵ_∞ and ϵ_0 are the susceptibilities in the infinite-frequency and steady-state limits respectively. Though 1-propanol was found to be well described by a simple Debye relationship, glycerol exhibited a higher degree of dispersion as vitrification accelerates and required unrealistically small values for its molecular volume in order for the observed relaxation time to conform to that predicted by the Debye hydrodynamic model for hard spheres. Davidson and Cole correctly surmised that this was due to the existence of additional modes of relaxation available to glycerol that simply cannot exist for the simpler and more rigid 1-propanol molecule. To deal with this anomalous dielectric dispersion, Davidson and Cole suggested modeling such associated systems with the empirically derived "skewed" Debye like model

$$\epsilon^* = \epsilon_\infty + \frac{\epsilon_0 - \epsilon_\infty}{(1 + i\omega\tau_0)^\beta}, \tag{4.2}$$

where β is the "skewing parameter." Note that Eq. 4.2 reduces to Eq. 4.1 when $\beta = 1$. Starting with Cole and Cole, it became common in the literature of this time to plot the real *vs*. imaginary components of this empirical expression as would have been common for the analysis of an analogous RC circuit[68]. With regards to complex susceptibility, in the context of dielectric spectroscopy, the real part of Eq. 4.2 corresponds to the non-dissipative component and the imaginary part to the dissipative component. This can be modeled, for a simple Debye relaxation where $\beta = 1$, in terms of a three component RC circuit as follows:

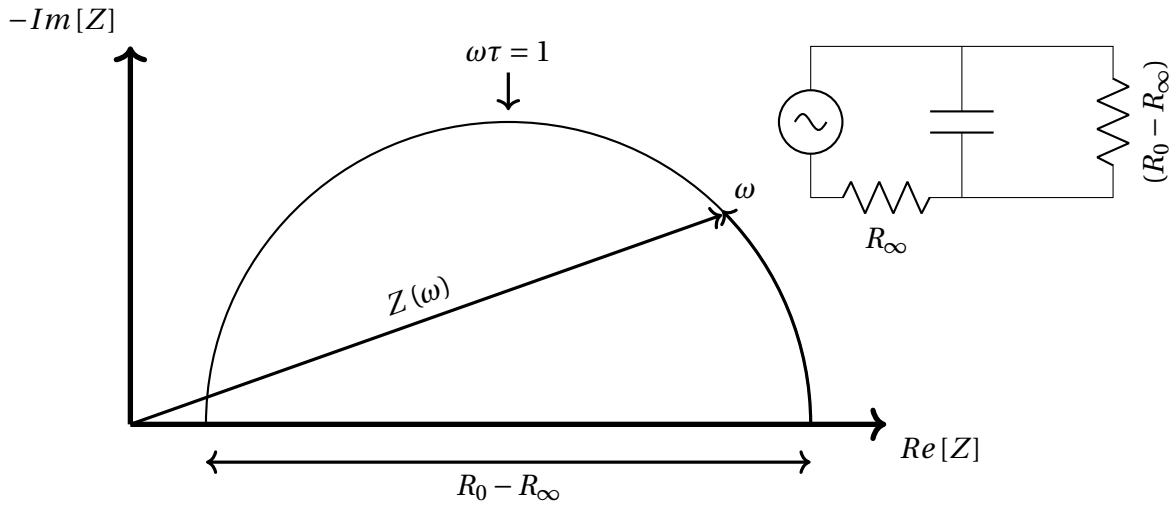


Figure 4.1: A system with a frequency dependent susceptibility and a single characteristic relaxation time τ driven at a given frequency ω can be modeled as an RC circuit as show here in the upper right corner. The complex impedance will be of the form $Z(\omega) = R_\infty + \frac{R_0 - R_\infty}{1+i\omega\tau}$ which is functionally identical to the dielectric susceptibility of the simple Debye relaxation of Eq. 4.1. Though the component which plays the role of irreversible dissipation is reversed in the case of the RC circuit above, both complex dielectric relaxation and this analogous electrical impedance will exhibit maximum dissipation at resonance where $\omega\tau = 1$.

Note that the magnitude of impedance will progress from R_0 to R_∞ as the driving frequency is increased from $\omega = 0$ to $\omega \to \infty$. The impedance of the two limiting cases, R_0 and R_∞, are analogous to the limiting values of the dielectric susceptibility ϵ_0 and ϵ_∞ from Eq. 4.1, respectively.

As previously mentioned, the Cole-Davidson model described by Eq. 4.2 above may be described as a "skewed" Debye model when $0 < \beta < 1$ as it results in a distortion of the corresponding Cole-Cole plot. An example of such a skewed plot of glycerol where $\beta = 0.6$ and n-propanol where $\beta \approx 1$, as obtained by Davidsion, is shown below. More importantly, Davidson provides a closed form expression for this distribution of simple Debye relaxations, first calculated by Fuoss and Kirkwood to describe dipole moments in polymers, which give rise to the Cole-Davidson model of Eq. 4.2 and this is described in more detail in Ch. 5 [69]. For now, we assert that this skewing of the dielectric dispersion, increasing as the value of β decreases, corresponds to a broader distribution

of Debye-like relaxation processes.

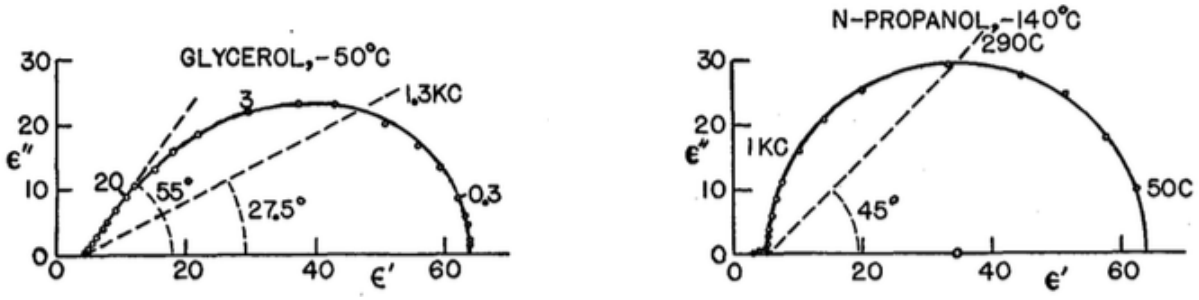


Figure 4.2: Cole-Cole plots of glycerol and n-propanol data (points). Solid lines represent fits to the Cole-Davidson model of Eq. 4.2 with beta-values of 0.6 and ≈ 1, respectively [49].

Davidson and Cole also noted that the value of β is essentially constant at ambient pressure over a temperature range from 40 °C to 75 °C. The implications of this being that the general structure of the distribution of relaxation modes is constant over a range of temperatures for which the dynamics of said relaxations are changing by at least six orders of magnitude. Similar results have been found for many systems over the years and for such systems, this is known as time-temperature superposition. More recently, the same has been found to hold over significant pressure ranges as well, in which case it is referred to as time-temperature-pressure superposition. We found such results for Cumene, are described in Ch. 5—β being essentially constant over a wide range of temperature and pressure.

Though Davidson and Cole only considered the isobaric behavior of glycerol relaxation as a function of temperature, their contribution here provided some of the first data that could be used to estimate fragility—at least at atmospheric pressure. They demonstrated that the strong temperature dependence of glycerol relaxation is poorly described, even over this moderate temperature range, by a classical Arrhenius rate equation but was well fit, within experimental uncertainty, by the more empirical Vogel-Tammann-Fulcher (VTF) equation previously described ~~in~~ Eq. 3.4

and first proposed by Tammann and Hesse to describe glycerol viscosity as a function of temperature [52]. Furthermore, they provided a closed form expression for the implied distribution of simple Debye relaxations underlying Cole-Davidson Eq. 4.2 above. Finally, they noted that the average molecular relaxation time of this distribution for glycerol would correspond to "hard-sphere" molecular dimensions, as provided by the classic Debye hydrodynamic theory, that are far too small to be physically realistic for glycerol, *e.g.* 1×10^{-25} cm^3, implying the existence of additional relaxation modes involving reorientation of molecular sub-units. It really is this observation, above all other contributions, that led to the development and application of free-volume models for viscosity and structural relaxation upon which much of this chapter will rely. Due to the importance of free-volume in this chapter and throughout the literature, particularly with regards to modeling viscosity of systems beyond ambient pressure, a discussion and derivation of the free-volume model will be provided in Sec. 4.2.

Throughout the 1960s, work began to emerge within the literature probing viscosity and structural relaxation of glycerol beyond ambient pressure. One of the first to appear was published in 1964 by McDuffie and Kelly where data for both viscosity, *via* a rolling ball viscometer, and relaxation data, *via* dielectric loss spectroscopy, were obtained. These methods were described previously in Sec. 2.3.3 and Sec. 2.3.5, respectively. McDuffie obtained viscosity data over three separate isotherms and dielectric relaxation data over a fourth which collectively covered a very small temperature range of $\Delta T = 20\,^{\circ}$C, however, ~~he~~ extended these isotherms over what was, at the time, a very large range of pressure from ambient up to 4 kbar. Beyond simply publishing some of the first high pressure data for glycerol viscosity, McDuffie's major contribution in this work was to show that neither the pressure nor temperature dependence of viscosity for glycerol was well described by the Eyring rate theory commonly used at the time and which will be described in the

following section. Also of considerable interest was that the free-volume model, ~~also~~ described in Sec. 4.2 below, was shown by McDuffie to yield a compressibility for glycerol that was not congruent with other high pressure values derived from ultrasonic probes at the time and also that the implementation of pressure-independent thermal expansion coefficients within the free-volume model leads to an over-estimation of the pressure dependence of viscosity. However these discrepancies were quickly addressed the following year, in 1966, by Matheson [70]. His arguably obvious approach was simply to apply a quadratic model to account for the pressure dependence of v_m in the free-volume expression (Eq. 4.29) from Sec. 4.2.1. An identical approach was applied for the temperature dependence for several other liquids however the limited temperature range in McDuffie's data was insufficient to meaningfully model a quadratic temperature dependence for v_m in glycerol. Still, in effectively addressing the shortcomings of early free-volume models exposed by McDufffie *via* extending the model beyond ambient pressure, Matheson's contribution was very significant. What Matheson was in want of, and what was conspicuously lacking in the literature up to this point, would be carefully collected data for glycerol viscosity or structural relaxation spanning a large temperature and pressure range.

This data vacuum was quickly filled by Johari and Whalley in the early 70's when they published a very heavily sighted paper containing dielectric susceptibility measurements of glycerol spanning a dynamic range of six decades. These data also contained a much larger pressure and temperature range than any previous work—ambient to 53 kbar and $-55\,^{\circ}$C to $84\,^{\circ}$C respectively [71]. At lower temperatures Johari and Whalley were able to probe the relaxation dynamics of glycerol very near the glass transition. At higher temperatures, above $T = 250$ K, their data falls around three decades short of approaching the glass transition. However their data has proven to be of superb quality and opened up the door to investigations of the pressure dependence of such

parameters as β, defined in Eq. 4.2, and much latter fragility. Similar contributions were made for several other glass-forming systems and many citations exist in the literature including this work, particularly to assess the pressure dependence of glycerol's fragility [72, 73, 74, 75].

Around a decade latter, in 1986, Forsman *et al*. published a paper also probing dielectric relaxation of glycerol [74]. They had the advantage of nearly a decade of advancement in instrumentation and thus were able to probe significantly faster relaxations, up to 13 MHz. However they did not provide a great deal of data and the pressure and temperature range was already well covered by Johari and Wally's previous work, particularly with regards to investigations into fragility as the faster relaxation times do not significantly add to the efficacy of fragility calculations. Though Forsman was focused more on the broadening of the loss peak as a function of pressure and furthers the discussion of many-body relaxation theory to account for deviations of the loss spectra from Debye theory, this work is worthy of mention here as it references several other papers providing data for glycerol relaxation collected over several decades.

Another work worthy of mention would be the work published by Demoulin in the early 70's [62]. Though data was only taken at ambient pressure and low temperatures, Demoulin probed glycerol relaxations directly in the time domain *via* photon correlation spectroscopy as described in Sec. 2.3.2. Though his data was confined to a relatively limited dynamic range, as autocorrelators of this era were among the first and significantly slower than those currently commercially available, with the aid of ultrasonic measurements a reasonable dynamic range was obtained. The aim of Demoulin was not to extend the dynamic range of data available for glycerol relaxation, but rather to prove the efficacy of digital photon correlation spectroscopy as a probe of relaxation dynamics over what was an exceedingly large dynamic range for the time. In Ch. 5, a method for extending this probe into the range of very high pressures will be presented. It is interesting here

to note that, though the data obtained *via* photon correlation spectroscopy is in the time domain, the Cole-Davidson distribution, being used pervasively throughout the literature to describe the dynamics of structural relaxation probed *via* dielectric spectroscopy and ultrasonics methods, is most naturally expressed in terms of frequency. This work was published before that of Lindsey and Patterson, which worked out a simple means of translating between the Williams-Watts distribution, expressed most naturally in the time domain and commonly used to model data obtained *via* photon correlation spectroscopy, and the Cole-Davidson distribution of Eq. 4.2 [63]. Thus it is not surprising that Demoulin chose to impart significant effort to fit his time domain data to the more common, yet inconvenient, Cole-Davidson distribution so that it may be directly compared to data within the literature. His method for accomplishing this will be extended in Ch. 5, taking advantage of modern computational tools, for exactly the same reasons.

Demoulin's method of PCS will later be well established to constitute a direct measurement of structural α-relaxation within a glass forming system. However another direct probe, this time of viscosity, was introduced before the end of the decade by Geils and Keezer [76]. The rolling ball viscometer as a means of directly measuring viscosity *via* tracking the terminal velocity of a metal ball rolling down an incline, as previously described in Sec. 2.3.3, certainly predated the work of Geils and Keezer, however early implementations placed great restrictions on the geometry of the viscometer requiring that the ratio of the rolling ball diameter to that of the viscometer remain in one or the other limiting extremes—either rolling without slipping down a surface of the viscometer or lacking any significant interaction with the confining surfaces [77, 78]. Geils and Keezer worked out the means of accounting for the correction factors associated with wall effects for this method, allowing for greater flexibility in experimental design to allow for temperature and potentially pressure control, drastically increasing its efficacy as a probe for viscosity of glass

forming systems.

4.2 Free-Volume Theory

The concept of free-volume as a mechanism to explain viscosity was first postulated as far back as 1913 when Batschinski proposed a linear relationship between thermal expansion and viscosity [79]. Though this linear model will obviously suffice over any sufficiently small thermodynamic range, it fails spectacularly in the description of data from high temperatures down through a glass transition. However the general idea of describing viscosity in terms of solid molecular units passing through regions of free-volume is certainly conceptually appealing and was gaining more traction in the literature in the early fifties. Though Batschinski's model was known by this time to be insufficient to describe viscosity behavior beyond a first approximation, and certainly would only be applicable far from any glass transition where viscosity is rapidly diverging, there was no reason to believe the concept of free-volume was invalid. The main competing theory of viscosity at the time was the rate theory proposed by Henry Eyring, which one may assert is physically equivalent to free-volume theory though couched in terms of the energy fluctuations associated with creating that free-volume—in the simplest terms, pushing nearest neighbors out of the way [80]. Eyring proposed that viscosity of any non-associated liquid, to the extent that structural relaxation is assumed to be an energetic process which involves breaking free of potential wells created by neighboring molecules, may be treated in much the same way as any chemical process activated by thermal energy fluctuations and should be well described by a Boltzman-like distribution involving a single activation energy, E_0, and of the following form:

$$\eta = \frac{Nd}{V}(2\pi m k_b T)^{\frac{1}{2}} e^{\frac{E_0}{k_b T}} \tag{4.3}$$

It is not surprising that an associated liquid such as glycerol might fail to be well described by this Arrhenius model. However even most non-associated fragile glass forming systems also conform well to the phenomenological VTF equation. Thus if one is to overcome the apparent discrepancy between Eyring's rate theory of Eq. 4.3 above and the VTF equation, the functional dependence of the activation energy on temperature must be of the form $E(T) \propto \frac{T}{T-T_\infty}$, where T_∞ is the zero mobility temperature at which viscosity becomes infinite. This requirement rather diminishes the physical grounding of the Eyring rate theory as it offers no explanation as to why the temperature dependence of activation energy should take this form. Some have tried to link the temperature dependence of activation energy with local entropy fluctuations and met with some level of success [81]. In fact such a rate theory is the foundation for the Adam-Gibbs equation which has been heavily implemented to describe enthalpy relaxation in many glass forming systems [82, 83]. However the conceptual simplicity and elegance of a free-volume description may suggest that a non-linear free-volume equation, beyond that suggested by Batschinski, may be the most physically meaningful approach to describing viscosity of these non-Arrhenius systems. It was during the same year of 1951 that Doolittle published his work describing viscosity of n-paraffin hydrocarbons *via* an exponential free-volume model, similar in form the the Eyring rate theory, yet dependent upon the relative free-space, $\frac{v_f}{v_0}$ within a bulk liquid rather than an activation energy,

$$\eta = Ae^{B\left(\frac{v_f}{v_0}\right)}, \tag{4.4}$$

where v_f is the volume of free-space per unit of liquid and v_0 is that free-volume density specifically evaluated at some reference temperature—often absolute zero [84].

This is useful as it circumvents the need to explain any functional dependence of a potentially complicated distribution of activation energies and attacks the problem from an arguably conceptually simpler standpoint of free-volume—is there sufficient free-volume, on average, for molecular mobility to progress at a given rate. Only thermal expansion needs to be considered as all other parameters are temperature independent. The Doolittle expression for free-volume of Eq. 4.4 above also provided the groundwork for Williams, Landel, and Ferry to explain the temperature dependence of kinetic constants as a function of temperature near the glass transition [85]. It was a major contribution that launched the free-volume model into common implementation. However Eyring's rate theory explanation was certainly not supplanted by a free-volume approach. Again, structural fluctuations that give rise to local free-volume, *i.e.* the opening of a path through which a caged molecule might diffuse, must be energetically accessible as must be the diffusion process itself. It should be unsurprising that a model relying on a temperature independent single activation energy to describe what, even in simple liquids, amounts to several independent modes of relaxation all of which may have strong temperature dependence, especially as the glass transition is approached, would prove to be every bit as insufficient as Batschinski's linear free-volume model. Eyring's model failed to sufficiently describe diffusion even in some simple crystalline systems [86].

Consider again the potential energy landscape of Fig. 2.6 way back in Ch. 2. Though, to some extent, structural relaxation may be entirely entropy driven as regions held at constant energy may still explore equipotential surfaces, moving from one local minimum within the energy landscape to another will be an energy activated process. Particularly as one approaches a state

here described as an ideal glass, the activation energy associated with hopping between adjacent potential wells will be strongly temperature dependent and the complexity of the parameter space implies there will be a near uniform distribution of potential paths through the parameter space corresponding to physical modes of structural relaxation. Thus Henry Eyring's description of viscosity as an energy activated process underlies the concept of free-volume. Bueche makes note, in his contemporary work on the segmental mobility of polymers, that one is left free to conceptualize relaxation processes in terms of the activation energy, ϵ, needed to break free of nearest neighbor cages or equivalently in terms of a activation volume, ΔV, needed for a molecular unit, or sub-molecular in the case of polymer chains, to pass through [87]. Bueche also importantly notes that, as previously alluded to by Davidson and Cole regarding the failure of the Debye hydrodynamic model to sufficiently explain the relaxation processes of glycerol, an activation volume, ΔV, need not be on the order of the molecular volume as relaxation modes may involve cooperative motions of sub-molecular units.

Bueche's desire to connect free-volume theory, which since the contribution of Doolittle had been shown to aptly model viscous behavior of many liquids over a much larger range of temperatures than when first proposed as a linear model by Batschinski, with the Eyring rate theory is quite understandable. Eyring's rate theory of viscosity, though experimentally found wanting, was strongly rooted in a first-principles derivation and shares an elegant symmetry with the well established thermodynamic theory of chemical reaction rates. If viscosity is, in fact, an energy activated process it should be governed by similar expressions to those governing chemical reactions. Though it may be convenient to model viscosity in terms of free-volume, Bueche argues that the creation of this free-volume comes at an energetic cost. The free-volume model proposed by Doolittle, though experimentally successful, falls short of the Eyring rate theory in its lack of any

theoretical explanation—the exponential form of Eq. 4.4 being entirely phenomenological.

A first-principles theoretical derivation of Doolittle's free-volume model was finally provided in 1959 *via* the work of Cohen and Turnbull [88]. To accomplish this, Cohen envisioned an entropic rather than energetic activation behind structural relaxation. Consider again the parameter space idealized by Fig. 2.6 as the n-dimensional surface that would exist for any real liquid. The energetic processes described by Bueche and Eyring may not be necessary to navigate a sufficiently complex parameter space. Couched in terms of the energy landscape picture, Cohen might suggest that one consider that what perhaps appears to be an insurmountable energy barrier between two adjacent potential wells in Fig. 2.6 in the absence of some introduced activation energy, ϵ, may in fact be accessible *via* multiple paths along equipotential surfaces in higher dimensional parameter space. Evolution along these equipotential surfaces correspond to pure entropy fluctuations at constant energy and can be physically interpreted as a rearrangement of existing free-volume within the bulk material.

Cohen and Turnbull do well to point out that the diffusion constant for spherical particles dispersed in a viscous medium is inversely related to viscosity *via* the Stokes-Einstein relation,

$$D = \frac{k_b T}{3\pi a_0 \eta}, \tag{4.5}$$

where a_0 is the hard sphere cross-section. Though this is derived in the context of low number density macroscopic spherical particles suspended in a viscous fluid, rather than the self diffusion of a pure liquid, it is still contextually relevant to the latter. In fact the use of nano-particles suspended in a liquid to serve as a probe of the molecular dynamics of the host liquid certainly appears in the literature and was considered as a possible method for extending photon correlation

spectroscopy into the diamond anvil cell, the results of which are described in Ch. 5. Suffice it to say for now that there was concern regarding the light arising from density fluctuations within a pure liquid from the very small volume of a diamond anvil cell being sufficient in intensity for photon correlation spectroscopy. As such it was considered that sufficiently small probe particles, perhaps within an order of magnitude of the molecular size of the liquid under study, might provide a sufficient scattering intensity and their diffusion would approximate, or at least track with, that of the molecular self diffusion for the host liquid.

This idea was in part inspired by the work of Caronna *et al*. who implemented x-ray scattering from suspended nano-particles to probe the dynamics of a glass forming liquid [89]. They found that at higher temperatures the particles undergo Brownian motion as predicated by Eq. 4.5 above however, as the glass transition is approached, the probe particles exhibit what the authors referred to as "hyperdiffusive behavior" which they suggest is due to the emergence of cooperative motions of the host liquid. It may be further suggested that, while approaching the glass transition, the effective size of the molecular units involved in structural relaxation increase accordingly as cooperative domains of the host liquid increase in size. This hyperdiffusive behavior will emerge as the size of these cooperative domains approach that of the probe particles.

The point to be had here is that, to the extent that a pure liquid can be modeled as consisting of hard spheres of radius a_0, self diffusion within a pure liquid should be well modeled by the Stokes-Eienstein relationship. Cohen and Turnbull assumed that this would hold true through the glass transition, however, as found much later by Caronna *et al*., this will depend on the emergence of cooperative domains that likely have the effect of imposing a strong temperature dependence on the effective size of the relaxing units, which may range from sub-molecular to macroscopic as the glass transition is approached. The emergence of cooperative domains and their roll regarding

structural relaxation within glass forming systems was not so well understood at this time. None the less, in the case of self diffusion from the perspective of free-volume, the effective size of the relaxing units, weather they be sub-molecular or large cooperative domains, will be intrinsically associated with the viscosity η. It is reasonable to assert that, no matter the functional form, to first approximation self diffusion within a pure liquid should be inversely proportional to the viscosity of the liquid.

The question that remains is weather this viscosity and diffusion result from energy activated processes or pure entropy fluctuations involving a rearrangement of free-volume within the bulk material. Cohen assumes the latter. What will follow here is Cohen's derivation of Doolittle's free-volume formula from first principles that exactly correlates with that of the energy activated rate theory of Eyring—the only difference being that energy constraints and statistical distributions are replaced with equivalent free-volume constraints and distributions. However it should perhaps be pointed out that, though Cohen here describes the process of diffusion, or equivalently viscosity, as arising from pure entropy fluctuations, the derivation that follows does not preclude free-volume being created or destroyed as the result of energetic processes. Generally speaking, a substance whose constituent molecules have sufficient average energy, $k_b T$, to exist in a liquid state will manifest local thermal fluctuations on the order of the heat of fusion giving rise to the creation of vacancies within the liquid—*i.e.* free-volume. Thus, arguably, one should not view free-volume theory as an alternative to an energy activated rate theory, but rather simply a more natural means of expressing such a theory *via* circumventing the need to parameterize the complicated distribution of temperature dependent activated processes that may give rise to the creation of free-volume.

The total free-volume of a bulk liquid, having average thermal energy $k_b T$, will undoubt-

edly fluctuate, however the expectation value of that total free-volume will be as solidly defined as any thermodynamic parameter. Cohen defines the average free-volume per molecule, analogous to the average thermal energy per molecule defined by Eyring, as $\nu_f = V_f/N$, where V_f is the total free-volume and N is the total number of particles in the bulk. Though it was reasonable for Eyring to assume the total energy of an isolated system is constant, there certainly will be fluctuations in the total free-volume, V_f, even in an isolated system, however such fluctuations will be of negligible magnitude and thus would likely prove inconsequential to any derivation. In any case such fluctuations remain irrelevant as V_f is either assumed to be the time-average value of the total free-volume or, assuming ergodicity, ν_f is simply defined as the ensemble average of local free-volume over the entirety of the bulk at any given instant. All of that being sufficiently addressed, we can move on to the derivation of Doolittle's free-volume model derived in a similar fashion as by Cohen.

4.2.1 Derivation of a Free Volume model

Let us consider a bulk fluid of finite volume. In general, there may be an infinite number of ways in which nearest neighbors to any particular molecule may be arranged within the fluid. However there is going to be a net free-volume, V_f, within the bulk at any given time which can be thought to be divided among N constituent molecular (or sub-molecular) units. Consider a set of small yet discrete increments of free-volume, $\Delta\nu$, that may be attributed to the ith molecule within the bulk as illustrated in Fig. 4.3.

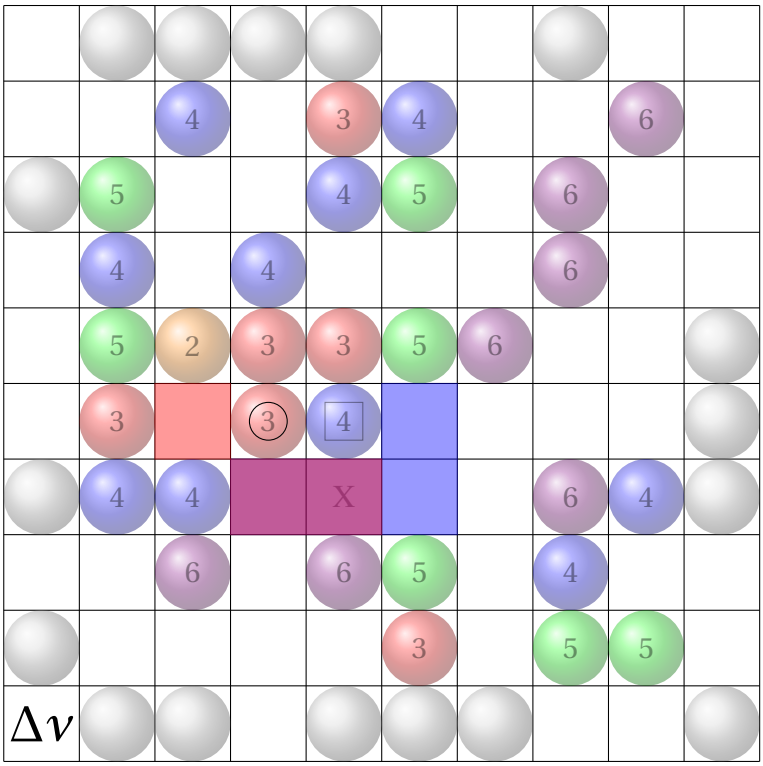


Figure 4.3: Here we have depicted a two dimensional conceptual illustration of free-volume in which the space is divided into small discrete volumes, Δv. Each particle is labeled with its number of associated free-volume increments, i, and color is used to indicate particles which share a common free-volume, v_i. The free-volumes associated with the circled and squared particles is indicated above *via* red and blue shading respectively where $v_\circ = 3\Delta v$ and $v_\square = 4\Delta v$. Thus, as asserted in Eq. 4.6, the total free-volume within the bulk will be given as $V_f = \gamma \sum n_i v_i$ where we are summing over all possible values of v_i, in this case $0 \leq v_i \leq 8\Delta v$. Note the need for the γ-factor to account for the overlapping free-volume indicated by the magenta region above. In this instance, $\gamma \approx 1/4$ implying that any given Δv will be shared with four adjacent particles on average, as happens to be the case for the magenta square marked with an X.

The ith molecule within the bulk may have an associated free-volume $v_i = i\Delta v$, where i is just an integer coefficient, such that $0 \leq v_i \leq V_f$ subject to the following constraints:

$$\begin{aligned} N &= \sum_i n_i \\ V_f &= \gamma \sum_i n_i v_i \end{aligned} \tag{4.6}$$

In the sum over all possible values of free-volume, v_i is the number of molecules having an as-

sociated free-volume of v_i, and $0.5 \leq \gamma \leq 1$ is a geometric factor to account for overlapping free-volume attributed to adjacent molecules. It should be noted that the lower limit of $\gamma = 0.5$ arises from the fact that a typical liquid will only share, on average, no more than half of its surrounding free-volume with at most one adjacent molecule. However, the illustration provided in Fig. 4.3 would, being excessively tightly packed, have a value of γ significantly less than one half. There is no physical reason why γ can not be less than one half and, in fact, for tightly packed granular systems almost certainly will be.

Getting back to our derivation, from here we will assume, similarly to Boltzmann, that any arrangement of free-volume among constituent molecules within the bulk is equally probable so long as the constraints of Eq. 4.6 above are met. Thus the odds of finding the free-volume distributed in any particular arrangement among N molecules such that n_i molecules have an associated free-volume v_i will be given by the total number of permutations of n_i as follows:

$$W = \frac{N!}{\prod_i (n_i!)} \tag{4.7}$$

From here Lagrange multipliers are implemented to find the most probable permutation. Just as is typical to derive the classic Boltzmann distribution we will, for mathematical convenience, maximize $\ln W$ invoking the Stirling approximation.

$$N! \approx \left(\frac{N}{e}\right)^N \tag{4.8}$$

Thus:

$$\begin{aligned} \ln[W] &= \ln[N!] - \sum_i \ln[n_i!] \\ &\approx N(\ln[N] - 1) - \sum_i \ln[n_i!] \end{aligned} \tag{4.9}$$

Note that we can now invoke the Stirling approximation on the second term of Eq. 4.9 resulting from the denominator in Eq. 4.7 and this will be done shortly. One might argue that there is no assurance the number of molecules, n_i, within the bulk having any arbitrary associated free-volume, v_i, is always sufficiently large as to justify the Stirling approximation. However, as is so often the case with thermodynamic quantities, the distribution of volumes, v_i, are distributed sharply around their average value. Therefor, though there certainly will be small values of n_i, neglecting them entirely will have negligible effect on the total free-volume of the system. Now, to maximize Eq. 4.9, we implement the Lagrange multipliers λ and μ as follows:

$$\delta \ln[W] - \lambda \delta N - \mu \delta V_f = 0 \tag{4.10}$$

First we consider $\delta \ln[W]$:

$$\begin{aligned} \delta \ln[W] &= \delta[N(\ln[N] - 1)] - \sum_i \delta[n_i(\ln[n_i] - 1)] \\ &= 0 - \sum_i \ln[n_i]\, \delta n_i - \sum_i \frac{n_i}{n_i} \delta n_i + \sum_i \delta n_i \\ &= -\sum_i \ln[n_i]\, \delta n_i \end{aligned} \tag{4.11}$$

Next we consider the constraints $\lambda\delta N$ and $\mu\delta V_f$:

$$\begin{aligned}\lambda\delta N &= \lambda\sum_i \delta n_i \\ \mu\delta V_f &= \mu\gamma\sum_i v_i\delta n_i\end{aligned} \tag{4.12}$$

Eq. 4.10 then becomes:

$$-\sum_i \left(\ln[n_i] + \lambda + \mu\gamma v_i\right)\delta n_i = 0 \tag{4.13}$$

As λ and μ are arbitrary Lagrange multipliers, asserting the sum to be zero implies that each term must be zero. From this we find the distribution of the number of particles, n_i, having associated free-volume, v_i which maximize the permutations of Eq. 4.7 corresponding the most probable distribution of free-volume within the bulk liquid as follows:

$$\begin{aligned}\ln[n_i] &= -\lambda - \mu\gamma v_i \\ n_i &= e^{-(\lambda+\mu\gamma v_i)}\end{aligned} \tag{4.14}$$

Thus the maximum permutations of Eq. 4.9 is found to be as follows:

$$\begin{aligned}\ln[W_{max}] &\approx N(\ln[N]-1) - \sum_i n_i(\ln[n_i]-1) \\ &= N(\ln[N]-1) - \sum_i e^{-(\lambda+\mu\gamma v_i)}\left(-\lambda-\mu\gamma v_i - 1\right) \\ &= N\ln[N] - N + \lambda\sum_i n_i + \mu\gamma\sum_i n_i v_i + \sum_i n_i \\ &= N\ln[N] + \lambda\sum_i n_i + \mu\gamma\sum_i n_i v_i\end{aligned} \tag{4.15}$$

Returning to the results of Eq. 4.14 and the constraints of Eq. 4.6 we can now express the total free-volume within a bulk liquid as follows:

$$\begin{aligned} V_f &= \gamma \sum_i n_i v_i \\ &= \gamma \sum_i v_i e^{-(\lambda+\mu\gamma v_i)} \end{aligned} \tag{4.16}$$

In the limit where the difference in free-volume among the discrete bins, thus far enumerated as v_i, becomes infinitesimal we may express the total free-volume of the bulk, V_f, as a continuous integral over an infinite number of infinitesimal bins.

$$V_f = \gamma \int_0^\infty v e^{-(\lambda+\mu\gamma v)} \delta v = \frac{e^{-\lambda}}{\mu^2 \gamma} \tag{4.17}$$

Solving this for the Lagrange multiplier, λ, yields the following:

$$\lambda = -\ln\left[\mu^2 \gamma V_f\right] \tag{4.18}$$

The constraints of Eq. 4.6 on the total number of molecules in the bulk can similarly be assessed.

$$\begin{aligned} N = \sum_i n_i &= \sum_i e^{\lambda+\mu\gamma v_i} \\ &= \int_0^\infty e^{-(\lambda+\mu\gamma v)} \delta v = \frac{e^{-\lambda}}{\mu\gamma} \end{aligned} \tag{4.19}$$

From here the average free energy per unit molecule, or the ensemble average $< v_f >$, is found to be as follows:

$$v_f = \frac{V_f}{N} = \frac{\left(\frac{e^{-\lambda}}{\mu^2\gamma}\right)}{\left(\frac{e^{-\lambda}}{\mu\gamma}\right)} = \frac{1}{\mu} \tag{4.20}$$

Returning to our previous solution for λ in Eq. 4.18 above we find the following:

$$\begin{aligned}\lambda &= -\ln\left[\mu^2 \gamma V_f\right] = -\ln\left[\frac{\gamma V_f}{v_f^2}\right] = -\ln\left[\frac{\gamma v_f N}{v_f^2}\right] \\ &= -\ln\left[\frac{\gamma}{v_f}\right] - \ln[N]\end{aligned} \tag{4.21}$$

In the limit where the difference in free-volume among the discrete bins becomes infinitesimal, we may express Eq. 4.14 as follows:

$$\begin{aligned}\delta n &= e^{-(\lambda+\mu\gamma v)} \\ &= \exp\left[-\left(-\ln\left[\frac{\gamma}{v_f}\right] - \ln[N] + \frac{\gamma v}{v_f}\right)\right] \\ &= N\left(\frac{\gamma}{v_f}\right) e^{\frac{-\gamma v}{v_f}}\end{aligned} \tag{4.22}$$

Thus we can express the number density of particles, $\rho(v)$, having an associated free-volume, v to be described as follows:

$$\rho(v) = \frac{\delta n}{N} = \left(\frac{\gamma}{v_f}\right) e^{-\frac{\gamma v}{v_f}} \tag{4.23}$$

where the normalization condition $\int_0^\infty \rho(v)\,\delta v = 1$ is naturally met. As previously mentioned, Cohen and Turnbull approaches this derivation assuming that self diffusion is not an energetically activated process, but rather is a purely structurally entropic process in which free-volume within the bulk can be thought of as being randomly redistributed *via* Brownian type motion along equipotential paths through the energy landscape. This is actually a reasonable assumption to the extent that a liquid is well approximated by a spherically symmetric force field where the intermolecular potential landscape is found to be relatively flat [90]. This assumption will hold true for almost any liquid in the high temperature limit where the potential energy landscape is not really felt by

the constituent molecules. At lower temperatures, certainly as one approaches the glass transition, energy activated process may play a larger role in the active creation of free-volume however the considerations of Cohen and Turnbull remain valid. A diffusion event is going to be inherently dependent on the existence of free-volume sufficient for molecular egress regardless of how that free-volume comes to be present. This makes free-volume arguably a more fundamental consideration in regards to molecular self diffusion than is any energetic rate theory. Thus ignoring for the moment how a constituent particle might find itself in the presence of free-volume v, it may be considered to be moving across a cage of nearest neighbors having a diameter of $\alpha(v)$ that is a function of its associated free-volume, v, and traveling with a velocity u. Cohen and Turnbull then reasonably assert that diffusion proceeds in much the same way as it does within crystals and any molecule's contribution to the diffusion constant will be given as follows [91]:

$$D(v) = \zeta u \alpha(v), \tag{4.24}$$

where ζ is just a geometric factor. As the distribution of free-volume is provided by the results of Eq. 4.23, the sum of all contributions will yield the following constant of self diffusion:

$$D = \int_{v_m}^{\infty} D(v)\, \rho(v)\, \delta v, \tag{4.25}$$

where v_m is the minimum value of free-volume needed to allow for any molecular egress and for which any contribution to diffusion can occur. The value of v_m is generally going to be at least an order of magnitude greater than the average available free volume per molecule, $< v_f >$. As such, Eq. 4.24 is going to be a slowly evolving function on the timescales of any particular diffusion

event. Thus Eq. 4.25 can be simplified as follows:

$$\begin{aligned} D &= \zeta u \alpha(v_m) \int_{v_m}^{\infty} \left(\frac{\gamma}{v_f}\right) e^{-\frac{\gamma v}{v_f}} \delta v \\ &= \zeta u \alpha(v_m) e^{-\frac{\gamma v_m}{v_f}} \end{aligned} \tag{4.26}$$

As Cohen points out, this is equivalent in form to the phenomenological model proposed by Doolittle [84]. In the case of "hard sphere" liquids, $\alpha(v_m)$ will be on the order of the spherical diameter and $u \propto \sqrt{k_b T}$ implies that even the simplest liquids must undergo a glass transition if cooled sufficiently in the absence of crystallization. Of course this model is certainly not limited to this $T^{1/2}$ temperature dependence. Eyring's general concept of a rate theory of diffusion, even extended to include any arbitrary temperature dependence or distribution of activation energies suggested by the broad applicability of the VTF equation, will here give rise to a temperature and pressure dependence of the average free-volume, v_f, related to the coefficient of thermal expansion. However, though energy activated processes may determine the average free-volume within a bulk, it is the existence of this free-volume, not the energy associated with its creation, that is fundamental to diffusion, viscosity, and relaxation processes. Thus Eq. 4.26 can be generalized as follows:

$$D(T,P) = \zeta u \alpha(v_m) e^{\frac{-\gamma v_m}{v(T,P) - v_m}} \tag{4.27}$$

The diffusion of spherical particles dispersed in a viscous medium described by Einstein and Stokes *via* Eq. 4.5 provided above can, to first approximation, be assumed to apply to self-diffusion of non-associated pure liquids well described by the flat potential energy landscape of hard spheres. It is physically reasonable to assume that, even when the potential energy landscape is significantly more complex, the diffusion rate will still remain proportional to fluidity or, equivalently, inversely

proportional to the viscosity. Thus one may arrive at the following simplified model for viscosity as a function of free-volume:

$$\eta(T,P) = C(T)\, e^{\frac{\gamma v_m}{v(T,P)-v_m}} \tag{4.28}$$

Here $C(T)$ is just some temperature dependent constant that will correspond to the viscosity at infinite volume which will be an exceedingly small value. It is worthy of note that viscosity really should trend to zero as free-volume approaches infinity. The fact that this model does not predict this implies that it cannot accurately describe viscosity over an infinite dynamic range regardless of the functional dependence of the parameters. However it does show remarkable agreement for most liquids over a very large dynamic range. As such it is common to expand this model around its value at ambient pressure as follows:

$$\eta(T,P) = \eta_0(T)\, e^{\frac{\gamma v_m}{v_m - v_0(T)}} e^{\frac{\gamma v_m}{v(T,P)-v_m}} \tag{4.29}$$

Here $\eta_0(T)$ and $v_0(T)$ are the temperature dependent viscosity and specific volume respectively at ambient pressure. As this model is generally applied to systems at or above ambient pressure, the physical incongruity regarding the non-zero infinite free-volume viscosity is not of significant concern. As implied by the super-exponential form of Eq. 4.29, these viscosities will increase by many orders of magnitude quite rapidly as either pressure is increased or temperature decreased. As such, it is very common to express Eq. 4.29 in logarithmic form as follows:

$$\ln\left[\frac{\eta(T,P)}{\eta_0(T)}\right] = \gamma v_m \left(\frac{1}{v(T,P)-v_m} - \frac{1}{v_0(T)-v_m}\right) \tag{4.30}$$

Some variation of Eq. 4.30 has been used extensively, with various functional dependencies of

parameters on temperature and pressure, throughout the literature to fit a large variety of associated and non-associated systems, including polymers, with high efficacy for viscosities ranging over several orders of magnitude [45, 92, 93, 94].

4.3 Reduction of extrapolation error in fragility measurements from pressure-limited data *via* implementation of TGP

Viscosity measurements at elevated pressure for glycerol up to about 4 kPa, first obtained by McDuffie and Kelly, was reported in the Journal of Chemical Physics in 1964 [95]. This pressure range was significantly extended by just under an order of magnitude *via* the work of Cook *et al*. in 1992 implementing a diamond anvil cell viscometer attached to a centrifuge [45, 51]. The data of Cook *et al*. were then used to investigate the pressure dependence of fragility over this domain and appeared to show a trend of steadily increasing fragility from approximately $m \approx 55$ ($m^* = 3.2$) at ambient pressure up to $m \approx 170$ ($m^* = 13.6$) at 3 GPa as shown in Fig. 4.4. Note that, though the definition of fragility provided by Eq. 3.2 is arguably more physically meaningful, it may be convenient for the sake of literature comparisons to express some fragilities in terms of the more classical definition of Eq. 3.1. Thus to alleviate any confusion we will adopt a consistent notation of m and m^* to indicate the classical fragility of Eq. 3.1 and the scaled fragility of Eq. 3.2 respectively. Returning now to Fig. 4.4, note the excessive error associated with these fragility calculations arising from the necessity to extrapolate the VTF model over several orders of magnitude up to the glass transition. The fact that this trend towards extreme fragility is not in satisfactory agreement with subsequent publications, particularly at elevated pressures, and the excessive error implying that the fragility measurements at high pressure are essentially meaningless has led to the Cook *et al*. data being generally under-emphasized. Paluch *et al*. drew attention to these concerns

as well as the fact that several of the viscosity measurements were made near the melt temperature, T_m, where VTF model extrapolations are known to be unreliable [73]. The fundamental limitation for Cook *et al*. was that rolling ball viscometery, even when implemented within a centrifuge, can only measure viscosities up to on the order of $10^9 cP$—*i.e*. as much as six orders of magnitude away from the glass transition at which fragility is assessed. The data of Cook *et al*., even with these concerns, does show strong support for an increasing trend in fragility with pressure up to around 2 GPa, which is not unexpected for a hydrogen bonding liquid where increasing fragility can be explained as arising from the diminished free-volume inhibiting hydrogen bonding. Beyond approximately 2 GPa, the minimal uncertainty in these fragilities obtained from propagating the errors in the fit parameters of the VTF equation seems to render the actual values of fragility statistically meaningless. Though the assessment of Cook *et al*., that the fragility continues to increase with pressure beyond this point, my have been somewhat compelling given that the measurements continue along the same well defined linear regression and their standard Angel plots, shown here as modified Angel plots and provided on the right side of Fig. 4.4, seem to consistently show increasing curvature as pressure increases, the higher pressure data is dubious to say the least, not just due to the excessive error but also in that, having reached a pressure of 3.0 GPa, glycerol has become about as fragile as a liquid can become. This linear trend in fragility certainly cannot continue indefinitely yet, at a pressure of 3 GPa, there is no indication of departing from said trend. Furthermore, it is worth noting that the 3 GPa fragility shown in Fig. 4.4 is based on only four data points and, though it is less evident in our modified Angel plot of Fig. 4.4 than it would be in a classical Angel plot, does not converge to a similar infinite temperature viscosity as do data obtained at lower pressures. It is therefor not surprising that this data might yield an excessively high fragility while having error bars that include negative values of fragility—statistically speaking this

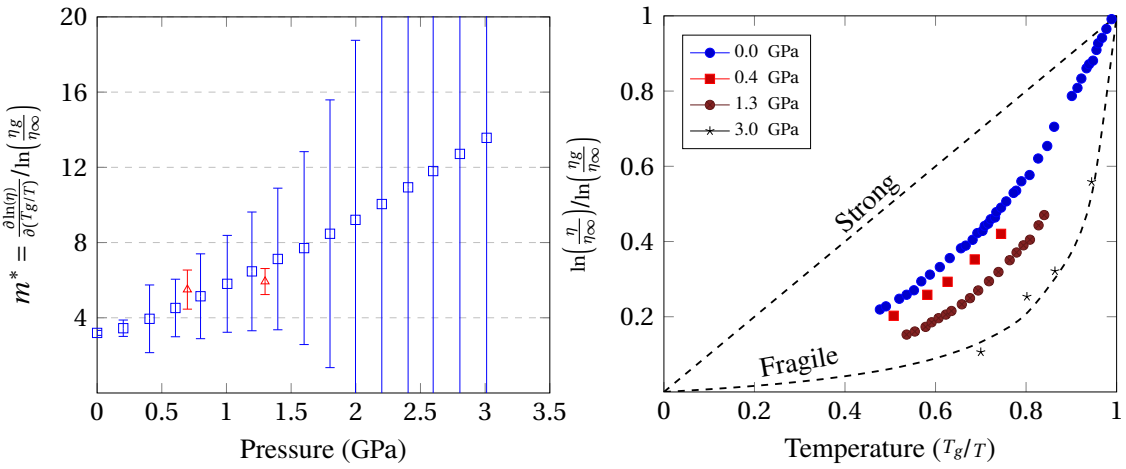


Figure 4.4: Left: Cook's calculations for the isobaric fragility of glycerol as a function of pressure with the glass transition defined at 10^{15} centipoise. Note that these values have been scaled to reflect our redefinition of fragility as defined in Eq. 3.2. This scaling of fragility is trivially accomplished as Cook conveniently provides his fit parameters. Right: Angel plots for Cook's 0.4, 1.3, and 3.0 GPa corrected isobars [51]. The 0.0 GPa isobar data is from Laughlin's 1969 Ph.D. thesis although all values plotted here were directly extracted from Cook's published figures [96].

is a meaningless measurement. In truth, this point should likely have been omitted from the analysis of Cook *et al*. as it relies on only three reasonably reliable data points—the 50.0 °C data point requiring a free-volume model extrapolation of almost 1 GPa. However omitting the 50.0 °C data point would leave only the 22.5, 75.0, and 125.0 °C, two of which still require slight extrapolation to 3 GPa, and require one to perform a non-linear model fit with only three data points providing constraint to a three parameter VTF model—though technically possible, an arguably meaningless endeavor. This was the root cause of their exploding fragility estimations—simply a lack of data to properly constrain the VTF model fits and extreme extrapolation over orders of magnitude as none of this data is within the vicinity of the glass transition.

More recent work has made a well supported case that the fragility of glycerol increases with pressure due to the inhibiting of hydrogen bonding up to approximately 2 GPa at which point it levels off becoming relatively constant. It would be reasonable to assert that the data of Cook *et*

al., as analyzed here, equally well supports this general trend of fragility. Arguably, the analysis supports a general increasing trend in fragility as pressure increases up to approximately 2 GPa even if little confidence can be put into the exact fragility values up to the point that the error begins to diverge rapidly. Beyond 2 GPa little can be confidently asserted from Cook's analysis however it does not contradict a constant fragility when the propagated errors are accounted for.

In order to assess the validity of Cook's high pressure data and the associated fragility, it is necessary to diminish these large error bars. This can be accomplished by addressing one of his two primary underlying issues—extreme data extrapolation. The TGP process developed by our group provides a direct measurement of the glass transition as a function of pressure allowing for a re-analysis of the Cook data without the need for extrapolation of the VTF model over so many orders of magnitude. This TGP data is provided in App. C (Tbl. B.1) along with the associated Anderson-Anderson regression fit. Using these values for the glass transition pressure on any given isotherm and assuming this transition to occur on an isochronal curve associated with a viscosity of $\eta = 10^{15} cps$, the data Cook *et al*. were refit to a classical VTF model.

Note that most of the data of Cook *et al*. were obtained over isotherms at varying pressures. In order to calculate fragility, they fit five sets of isothermal viscosity data to the classic free-volume model of Eq. 4.30. The Tait equation of state, fit to density data provided by Bridgman, was implemented to convert pressure to volume [97]. More details regarding the implementation of the Tait equation of state for glycerol are provided in App. E as Cook *et al*. and this work use this free-volume model to obtain (mostly) interpolated isobaric data as true isobaric data is difficult to obtain experimentally given that pressures will vary significantly within the DAC under thermal expansion. Thus it is this interpolated isobaric data that is used in the fits to the VTF model for fragility analysis.

Fig. 4.5 (left) provides a scaled Angel plot both excluding the TGP glass transition point (solid) as well as the refit passing through the TGP point (dashed) for a 1.3 GPa isobar. Note that the lower five data points associated with both the dashed and solid regressions are in fact the same data from Cook *et al*. —scaled according to their associated VTF fits and glass transition temperatures. The TGP point is also provided and is, on a scaled Angel plot, located at (0,0) by definition. It is immediately clear that both adequately describe Cook's data however omitting the TGP data leads to a significant over-estimation of fragility. This is associated with an over-estimation of the glass transition temperature. Fig. 4.5 (right) shows eight isobaric data sets, consisting of four to five data points each, ranging left to right from 0 GPa to 3 GPa which are obtained from the before mentioned free-volume fits to Cook's published isothermal data. The VTF fits to this data exclude the TGP point and thus help to demonstrate the error that arises from the extrapolation of the VTF equation. Notice that at the lowest pressures (left most curves) there is a systematic under-estimation of the glass transition temperature due to model extrapolation. However, as we move up in pressure (to the right of the figure) we begin to increasingly over-estimate the glass transition temperature and thus over-estimate the fragility. This trend is what is giving rise to Cook's exploding fragility measurements.

From this reassessment of the data of Cook *et al*., we see a very different behavior of fragility as a function of pressure. Though, as demonstrated in Fig. 4.7, the calculations of fragility attained from the Cook *et al*. data ~~enhanced *via* the~~ TGP isochronal data were still not found to be in ~~perfect~~ agreement with more recent measurements, the general trend is found to be somewhat congruent [66, 98]. It should be noted that the actual values of fragility, apart from their trend with pressure, ~~is~~ somewhat dependent on the viscosity at the glass transition. We have chosen to assume the TGP transition correlates with a viscosity of $\approx 10^{15} cps$. A quick assessment of Fig. 4.6

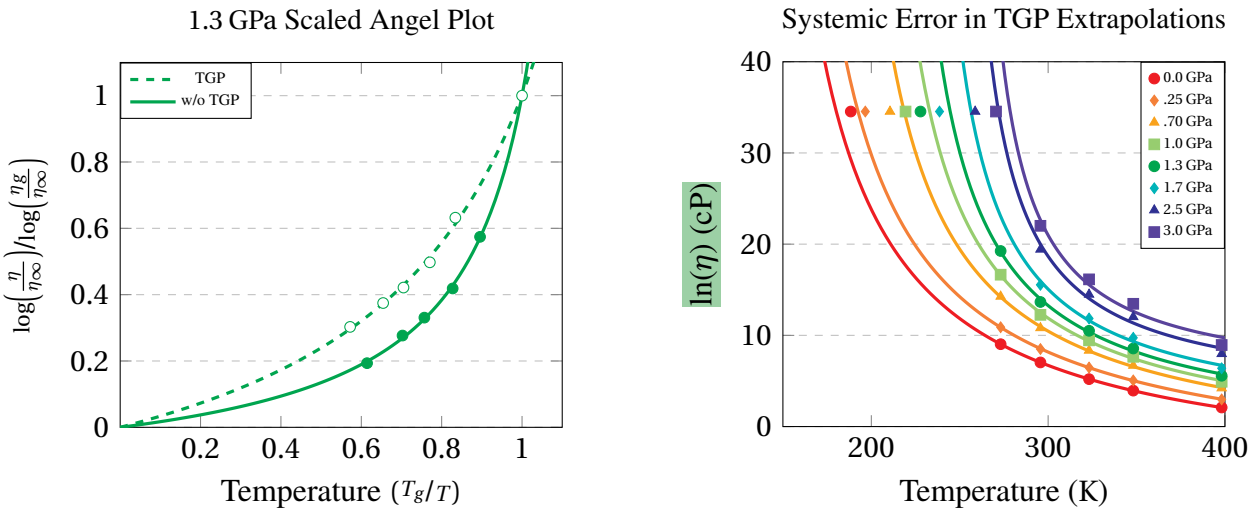


Figure 4.5: Left: A scaled Angel plot of the data of Cook *et al*. clearly demonstrates the over-estimation of fragility occurring at 1.3 GPa due to extreme extrapolation up to the glass transition that is rectified *via* the inclusion of TGP isochronal data. The same data is provided for both the solid and dotted VTF fits—scaled accordingly to their fit parameters and the glass transition temperatures they predict. Right: Isobaric data from Cook *et al*. showing the general trend of under-estimation followed by increasing over-estimation of the glass transition temperature with increasing pressure. The TGP method provides the isochronal data at the glass transition ~~[51]~~.

might lead one to conclude that fragility is initially decreasing from ambient pressure up to approximately 1.5 GPa however, as evidenced by the error of the VTF fit parameters propagated through the fragility calculation as well as keeping in mind that our confidence in these values is still limited *via* insufficient data particularly in the region of the glass transition, this trend should not be assumed statistically significant—the fragility is relatively constant up to around 1.3 GPa. It is worth noting here that, though ~~Cook's~~ viscosity data was predominately taken along isotherms as this was experimentally favorable, he did manage to obtain two pseudo-isobaric runs at 0.7 and 1.3 GPa on which ~~he~~ then used interpolation between free-volume fits to his isothermal data to correct for slight deviations of pressure. Though this process was extremely time intensive, it did provide ~~him~~ with two isobaric runs over a large temperature range addressing the issue of insufficient data points. It should be noted, however, that these data are still limited to well below the glass transi-

tion. Due to the larger number of data points associated with these pseudo-isobaric runs, the error associated with the fragility estimations are significantly smaller and these two points are included (△) in the left side of Fig. 4.4. These two higher-confidence points would alone imply a potentially constant fragility below 1.5 GPa—albeit at a significantly higher scaled fragility ($m^* \approx 5.7$). Beyond 1.5 GPa the fragility steadily increases by $\Delta m^* \approx 7.8$ and then remain approximately constant within uncertainty beyond 2.5 to 3 GPa before rapidly diverging due to extrapolation errors again. This general trend of increasing fragility is, in the loosest sense, in "agreement" with subsequent published works as will be demonstrated. This leads to the conclusion that the viscosity data of Cook *et al*. is perhaps significantly more reliable than may have been assumed and much of its error, as perhaps expected, is predominately the result of these extrapolation issues. Our method of employing TGP data to diminish this extrapolation error significantly decreases the error associated with these fragility estimations however it is still limited—particularly at elevated pressures. More data is needed in the region of interest to further define the slope at the glass transition. It is possible that additional TGP runs with longer or shorter annealing periods could provide this however in the absence of a solid equation of state at high pressure and thermodynamic scaling to provide the effective viscosity at the glass transition to higher precision than the assumed 10^{15}cP, this is not yet a viable means of defining the slope near the glass transition. Other methods such as photon correlation spectroscopy and dielectric loss spectroscopy certainly exist and there is no shortage of data for glycerol dynamics within the literature around the vicinity of the glass transition, though generally limited to cryogenic measurements at low pressures. If the issues with these fragility calculations do in fact stem predominately from the lack of data in critical regions rather than the efficacy of the data itself, then supplementation of their data with other literature data in critical regions should allow accurate determinations of fragility as a function of pressure. Glyc-

erol data taken by several different groups extend over different pressure and temperature regions and include both viscosity and α-relaxation measurements, making glycerol a prime candidate for a surface fragility analysis, similar to what will also be done for cumene in Ch. 5.

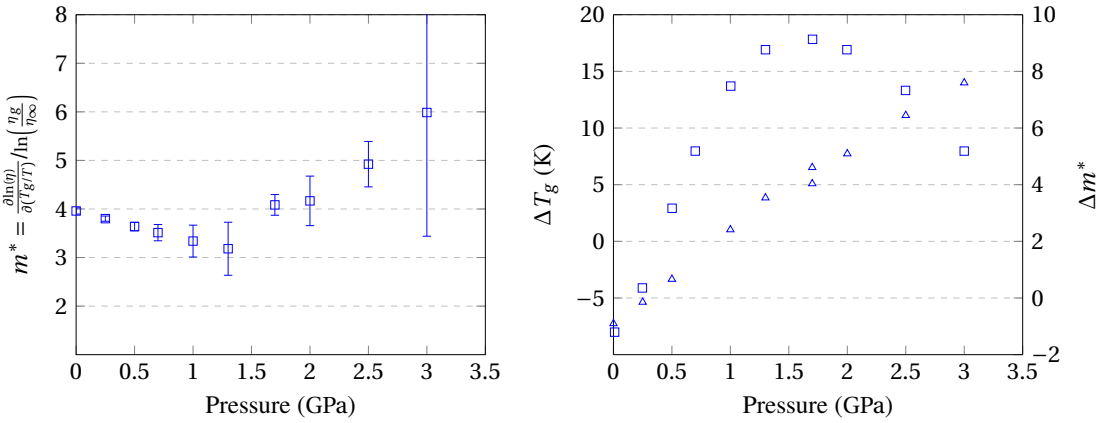


Figure 4.6: Left: Scaled fragility as a function of pressure derived from classic VTF fits to the data of Cook *et al*. with the glass transition point included as determined *via* our TGP isochron. Right: The error in terms of under or over-estimation of the glass transition temperature, ΔT_g, due to extreme extrapolation of the VTF model as demonstrated in Fig. 4.5, is provided (□) along with error in scaled fragility (△) calculated from these extrapolated models.

The right side of Fig. 4.6 shows the under or over estimation of the glass transition temperature as a function of pressure derived from the classic VTF fits to the data of Cook *et al*. without the TGP isochronal data. The initial under-estimation of the glass transition temperature for the two lowest pressures is followed by a trend of increasing over-estimation up to about 1.7 GPa where we continue to over-estimate the transition temperature, albeit to an increasingly lesser degree. However the error in fragility associated with this over-estimation in glass transition temperature continues to increase in a fairly linear fashion leading to the excessively high fragilities at increased pressure observed by Cook *et al*. in Fig. 4.4.

To add further validation to Cooks data at lower pressures, the same analysis method of implementing the TGP data point to constrain the VTF fits in the vicinity of the glass transition was

applied to the rolling ball data obtained by McDuffie and Kelly and the associated fragilities calculated [95]. Though these fragility values were much more in line with other works implementing dynamic light scattering (Brodin [99]), and dielectric spectroscopy (Reiser [98], Pronin [66], and McDuffie [95]), they all follow the same general trend of constant fragility below 1 GPa. The increasing trend mentioned by Pronin and Reiser is simply not statistically significant. This seems to suggest a strong increase in fragility does not occur until beyond 1 GPa. This may be due to the general expected trend of decreasing fragility with increasing pressure is being approximately balanced by the early onset of hydrogen bond breaking giving rise to a relatively constant fragility in this range.

4.4 Structural Relaxation Surface

Experimentally, it is often difficult to obtain truly isobaric data at high pressure. When a fluid is confined within a pressure vessel or diamond–anvil cell, changes in temperature inevitably induce changes in pressure due to thermal expansion. In his PCS measurements, Cook went to considerable lengths to interpolate between data points in order to construct pseudo–isobaric data runs. Of considerable utility, therefore, would be a framework capable of fitting relaxation data acquired over broad ranges of temperature and pressure (or specific volume) without requiring strict isobaric or, for that matter isothermal, constraints. This motivates the development of a continuous structural–relaxation surface.

To this end, the Andersson–Andersson constrained surface model provided in App. A is employed, extending the classic isobaric Vogel–Tammann–Fulcher (VTF) formalism to account for pressure dependence through polynomial expansions of pressure–dependent parameters. A more rigorous mathematical development of this model is presented in Sec. 5.4.2. Here, only the

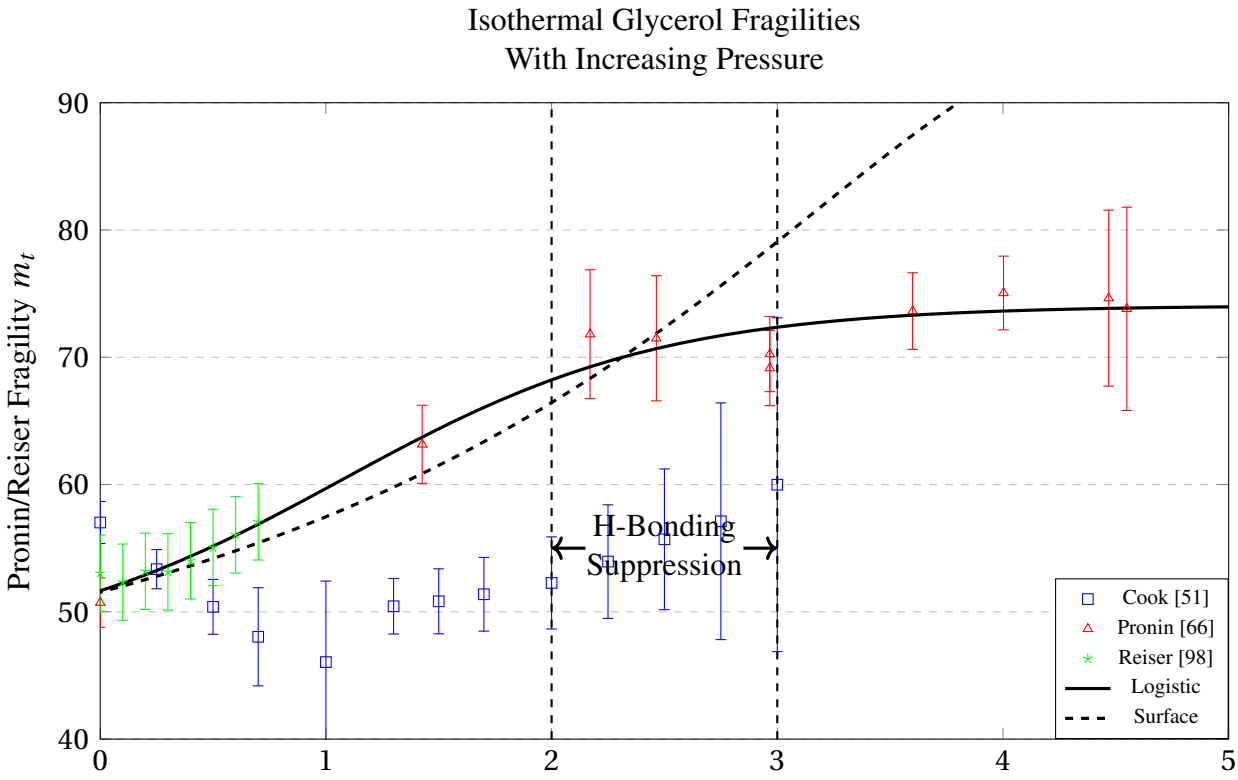


Figure 4.7: The fragility measurements calculated from classic VTF fits to Cook's viscosity data corrected *via* the inclusion of our TGP glass transition point (□) is shown above to be no longer wholly incongruent with more recent work. A Logistic fit to Pronin (△) and Reiser's (⋆) data describes their fragility measurements well with the increase in fragility seeming to saturate between 2 and 3 GPa as originally asserted by Paluch [73] and Pawlus [75]. The Cook fragilities retain relatively large uncertainties, thus cannot reflect accurately the fine behavior of fragility as a function of pressure, yet do seem to reflect this fragility increase between 2 and 3 GPa. A reasonable trend-line is provided that falls within all error bars.

essential structure of the surface and the physical constraints imposed upon it are discussed.

For most liquids, a substantial body of high–quality ambient–pressure data exists in the literature, obtained using a variety of experimental probes of structural relaxation and viscosity, owing to the relative experimental accessibility of this thermodynamic region—glycerol being no exception. Fig. A presents an Angell plot of all atmospheric–pressure data collected for this work, together with a fit to the classic isobaric VTF model. This atmospheric isobar serves as a primary constraint for the relaxation surface.

It should be noted that the Cook data have been excluded from this atmospheric–pressure constraint. The TGP point, shown on the same Angell plot at a location corresponding to $\tau = 220$ s as determined from thermodynamic scaling of the low–pressure TGP data in Fig. 4.5, was likewise not included in the fit, though it exhibits excellent agreement with the remaining data.

At the time the atmospheric data were fit, the degree to which the intercept method of thermodynamic scaling accurately determines the isochronal value associated with the Andersson–Andersson regression of the TGP data was not yet established. One peripheral insight arising from the development of these surface models is the realization that, for liquids whose dynamics conform to thermodynamic scaling, this approach provides a robust means of identifying the precise isochrone represented by a given TGP data run. Even in cases where thermodynamic scaling may not strictly apply, the atmospheric–pressure constraint provides an independent verification of this isochronal value, as the Andersson–Andersson regression through the TGP data must intersect the atmospheric VTF isobar.

The Andersson–Andersson regression through the TGP data provides an additional constraint to be imposed on the surface construction. Specifically, the VTF expression is solved for the zero–mobility temperature T_0 in terms of the pressure–dependent glass–transition temperature $T_g(P)$ and subsequently reinserted into the VTF form, yielding Eq. A.2. Here, $T_g(P)$ is taken from the Andersson–Andersson fit to the TGP data provided in App. B. This procedure results in a relaxation surface constrained both at atmospheric pressure and along an isochrone corresponding to $\tau \approx 220$ s.

The remaining four parameters comprising the coefficient vectors $\vec{d}$ and $\vec{t}$ are treated as free parameters and are determined using a Levenberg–Marquardt optimization algorithm to conform the surface to the remaining data. Fig. 4.8 displays this constrained surface together with the Cook

data, shown for initial parameter estimates. The atmospheric isobaric constraint and the isochronal TGP constraint are indicated by solid black curves on the surface.

It is emphasized that the surface is not fit to the Cook data. Because Cook measured viscosity directly via rolling–ball viscometry, whereas all other data in this chapter probe the structural α–relaxation time, the decision was made to include the Cook data only for qualitative comparison. The Cook viscosity data were scaled using an appropriate high–frequency shear modulus G_∞ to permit comparison with the relaxation–time surface. The color scale in Fig. 4.8 indicates pressure and demonstrates that the overall pressure–dependent trends of the Cook data are well captured by the constrained surface.

**Next we move on an clean what follows up. We add tau data, fit the surface, and again the cook data is NOT included in the fit...just placed on the surface

Glycerol viscosity and relaxation data from several sources are shown, along with the Cook rolling ball data obtain from within a DAC, in Fig. 4.9 [99, 95, 73, 66, 98]. It is clear that data is actually in excellent agreement with other literature data further bolstering the case that the vast majority of the error within the Cook fragility measurements lies in the extreme extrapolation to the glass transition. The inclusion of the TGP isochronal data, though addressing the issue of extrapolation to a significant extent, ultimately provides insufficient accommodation for an accurate determination of the rate of vitrification with increasing pressure in the vicinity of the glass transition, T_g. More data is needed to define the slope near T_g to obtain an isobaric fragility in consensus with other contributors to the literature, *i.e.* Pronin, Paluch, *etc*.

In short, there never was any issue with the validity of Cook's data. As we will see, it is in excellent agreement with other measurements of vitrification or associated alpha relaxation behavior. It is only the lack of data to provide constraint to the VTF fit in key locations, specifically

in the vicinity of T_g, that gives rise to the incongruence of the Cook fragility estimations with literature values, *i.e.* the slope in the vicinity of T_g is poorly constrained. The addition of the TGP isochronal data is not itself sufficient to provide the necessary constraint on this slope for reliable estimations of fragility. However, as we will here demonstrate, only one additional experimentally trivial, constraint is needed for Cook's data to provide fragility values as a function of pressure in excellent agreement with the current literature consensus—an ambient isobar.

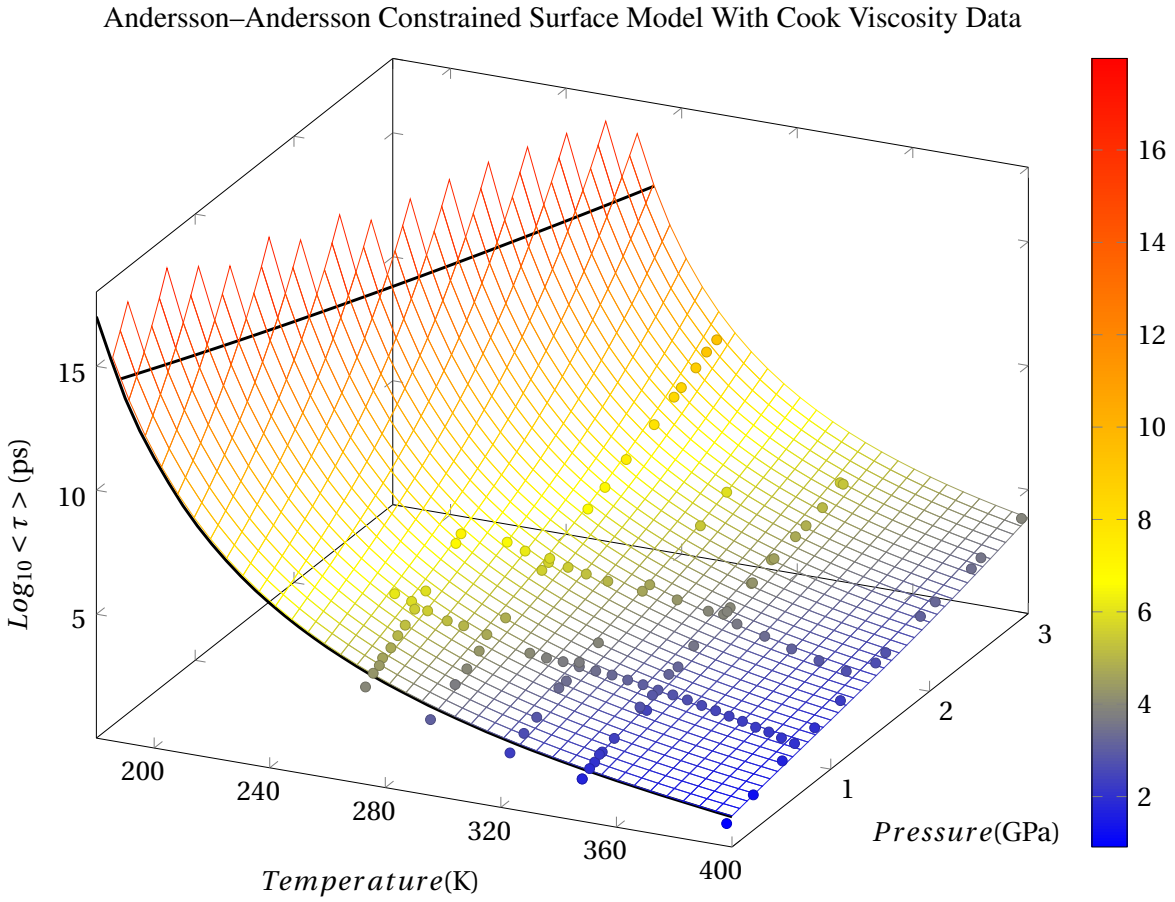


Figure 4.8: Cook's rolling ball viscosity data is here provided along with the Andersson-Andersson constrained surface model, given initial approximations for the fit parameters, described in App. A. The color scale, corresponding to pressure, facilitates visualization of how well the data conforms to the surface. The surface is extended to 3 GPa corresponding to the highest pressures for which Cook obtained data [51].

Fig. 4.8 shows all of Cook's raw viscosity data provided by reference [51] fit to the Andersson-

Andersson constrained surface model described in App. A. The Solid black isobaric line at atmospheric pressure, through which the surface is constrained, is well described in Fig. A, along with the supporting data. This 1-bar regression was also used to define the isochrone for the TGP curve—the second solid black line extending across the top of the surface in Fig. 4.8. The glycerol TGP data fit to a standard Andersson-Andersson model is provided in App. B and corresponds to an isochronal pressure-temperature relationship at $\tau \approx 10^{14}$ s through which the surface is also constrained.

This Andersson-Andersson constrained surface model, fit to the Cook viscosity data, allows for a determination of fragility as a function of pressure in similar fashion to that obtained for cumene in Fig. 5.16 previously. The resulting functional dependence of isobaric fragility is plotted as a solid black line in Fig. 4.7 above and found to be in significantly greater agreement with Pronin, Reiser, and other more recent works up to 2.5 GPa. This result, along with the excellent constrained surface fit, suggest that Cook's data could not be lacking significantly in quality.

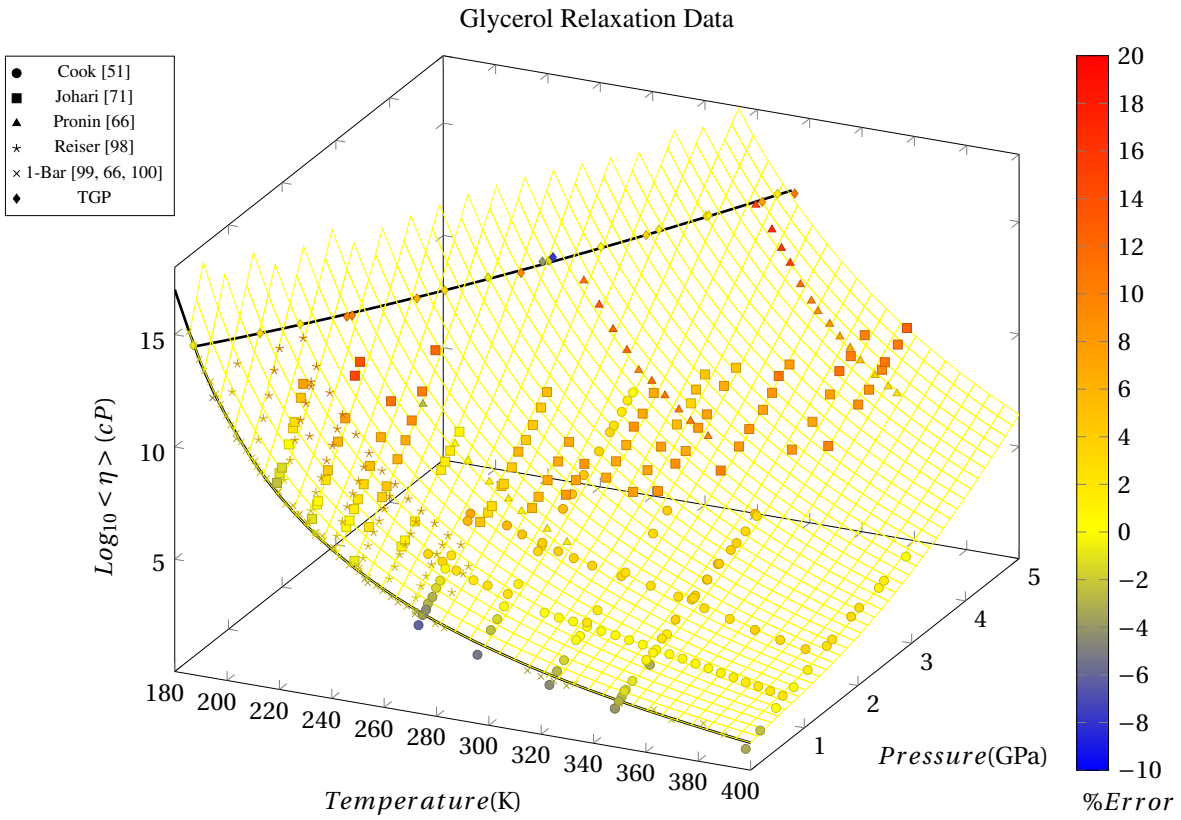


Figure 4.9: The surface shown corresponds to the model described in App. A, fit to a compilation of glycerol relaxation data from Johari and Whalley, Pronin, and Reiser. The Cook, TGP, and atmospheric data were not included in the fit, but are represented on the surface. Data points are colored by their percent deviation from the fitted surface, highlighting systematic trends in the residuals.

Here we have the model described in App. A fit to a large selection of glycerol relaxation data from multiple contributers [71, 66, 98]. The same TGP and 1-bar constraint was imposed here as was implemented in Fig. 4.8. The color scale now represents percent deviation of the data from the surface. Note how the high pressure Pronin data begins to diverge from the surface, approaching an error as high as 20%. This data was still included in the fit as it conforms well to the surface at higher temperatures and lower viscosities—only tending away from the surface in a region where the TGP constraint does not allow the data significant influence over the fit parameters.

Fig. 4.8 shows the same Andersson-Andersson constrained surface model fit to a larger selection of literature data including Cook's rolling ball viscosity data. The same TGP and atmospheric constraints are applied. Here the Actual TGP data is provided on the surface yet not included in the fit. It is found that Cook's rolling ball viscosity data is in excellent agreement with other works including Johari, Pronin, Paluch, and Reiser thus we must conclude that the Cook data is very reliable—the issue of increasingly overestimated fragility arising predominately from extreme extrapolation [29, 73, 66, 98]. It should be noted that the four isobaric data sets extracted from Pronin's Fig. 3 in reference [66] do trend increasingly above the fit surface at higher pressures yet agree with Cook and others at lower pressures. The toroidal pressure cell used by Pronin and described in reference [101] may not have provided a hydrostatic sample to the degree that was assumed—a scenario which would certainly explain the data's increasing divergence from the TGP isochrone at higher pressures and lower temperatures as the glass transition is approached. The Pronin isobars are still in reasonable agreement with Cook's data at higher temperatures where hydrostatic conditions are assured, even at the highest pressure. However this data was included as it does not heavily effect the surface at lower pressures where it is in reasonable agreement with the rest of the data yet allows us to extend this fit surface up to 5 GPa. Thus it should be assumed that our calculated fragility above 3 GPa may have reasonable error associated with it and any fine structure is not of high confidence. It should be noted that this Pronin data could be replaced with DPCS measurements at high pressure near the glass transition just as was done with cumene. Molecular simulations may provide high pressure fast dynamics at extreme temperatures.

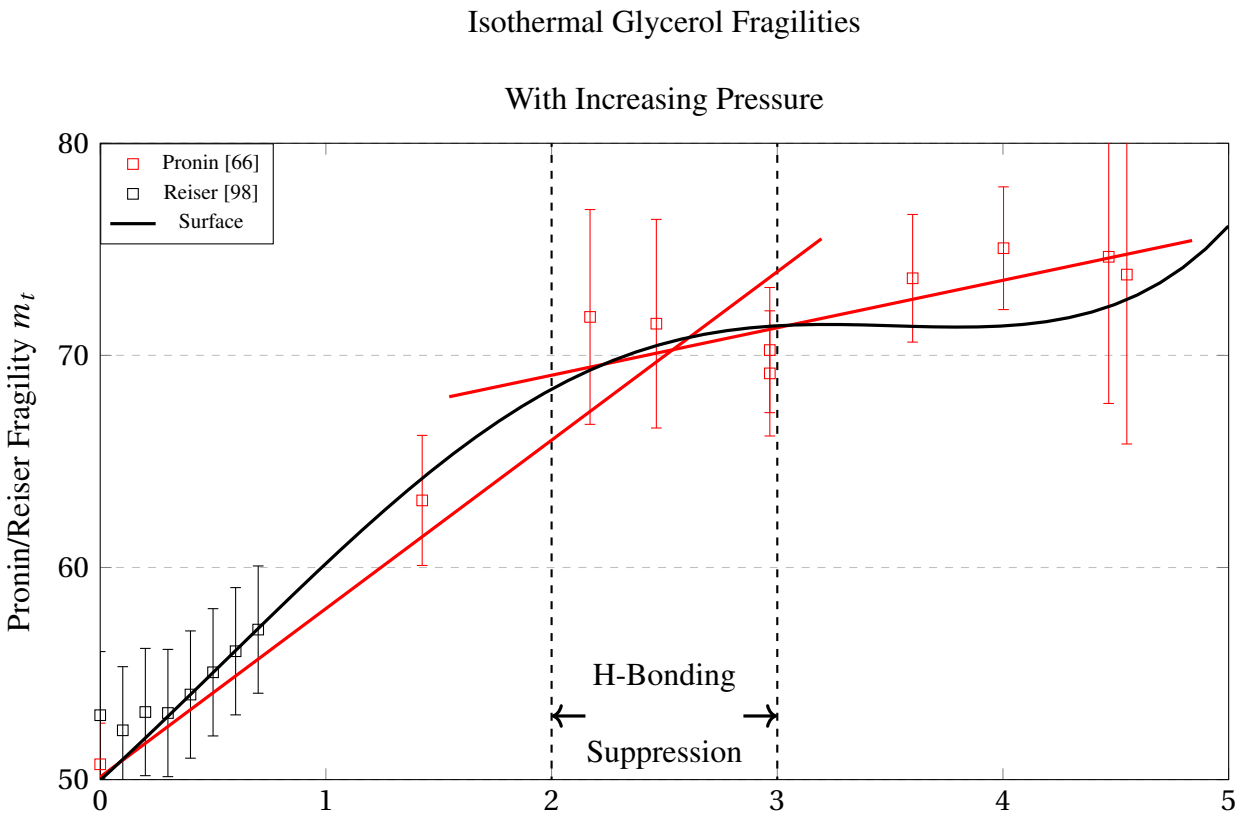


Figure 4.10: Fragility estimations from Pronin (□) and Reiser's (□) data are shown above. The pressure dependence of fragility derived from isobaric temperature derivatives of the Andersson-Andersson surface model described in App. A, fit to the data of Cook, Johari, Pronin, Paluch, and Reiser, is shown as a solid black line. The upward curve in fragility approaching 5 GPa is likely a relic due to the previously discussed issues with the Pronin data at high pressure as well as the surface being generally unconstrained by data beyond this pressure.

The calculated fragility as a function of pressure arising from the Andersson-Andersson surface model from Fig. 4.9 including data from Cook, Johari, Pronin, Paluch and Reiser is provided in Fig. 4.4—extending our fragility measurements out to 5 GPa. The red lines show Pronin's original estimation of the fragility behavior with increasing pressure as was lifted directly from his Fig. 4 in reference [66]. This result shows unequivocally Cook's rolling ball viscosity data to be in excellent agreement with other literature measurements of viscosity, η, and α-relaxation, τ. The Andersson-Andersson constrained surface model relieves us of the experimental burden of

maintaining isobaric conditions during data acquisition, something that is very difficult to accomplish in a diamond anvil cell, as well as avoids the reliance on free-volume models where a reliable equation of state may not be available or more dubious isothermal models such as the modified VTF equation.

4.5 Equation of State and Thermodynamic Scaling

In Ch. 5 Sec. 5.4.1 it is shown that the relaxation dynamics of a simple Van der Waals liquid such as cumene conforms very nicely to a single function of the form $\tau \propto \frac{1}{TV^{\gamma}}$, where γ is a constant scaling factor. This factor gamma has been shown to depend on the intermolecular potential energy landscape, $U(r)\sim\left(\frac{\sigma}{r}\right)^{3\gamma}$, where r is the spatial separation between any two molecules and σ is a material constant [102, 103]. In the case of a simple liquid such as cumene the potential is well approximated by the form of the Lennard-Jones model for which $3\gamma \approx 12$—consistent with the experimentally determined value of $\gamma \cong 4.77$. The dynamics such liquids are strongly driven by simple free-volume considerations and thus we expect thermodynamic scaling to hold with increasing pressure and that is what we have observed to be the case.

However in the case of strongly associated liquids such as glycerol, intra-molecular bonds influence the potential energy landscape. As these bonds are dependent on temperature and pressure, the energy landscape will be temperature and pressure dependent, thus one would not expect the emergence of thermodynamic scaling in these liquids. However several such liquids do exhibit thermodynamic scaling—at least over limited ranges of pressure and temperature. Glycerol is one such example, having a measured scaling parameter within the literature in the range of $1.4 \leq \gamma \leq 1.6$ [98, 104]. This lower value for γ is expected given that free-volume will play a reduced role in the dynamics of associated liquids as molecular mobility is no longer just a mater

of thermal diffusion past nearest neighbors exhibiting Van der Waals repulsion, but also will require energy to break hydrogen bonds with said neighbors. However these values for γ were all determined at relatively low pressures—on the order of 0.1 MPa. Yet we see in Fig. 4.4 a shift in the fragility behavior as a function of pressure occurring between 2 and 3 GPa which has been attributed to a suppression of hydrogen bonding. If hydrogen bonding between glycerol molecules is sufficiently suppressed at very high pressures, one might expect glycerol to behave much more like a simple Van der Waals liquid with a scaling parameter, γ, much closer to 4 as the potential energy landscape no longer is heavily influenced by these suppressed intra-molecular interactions.

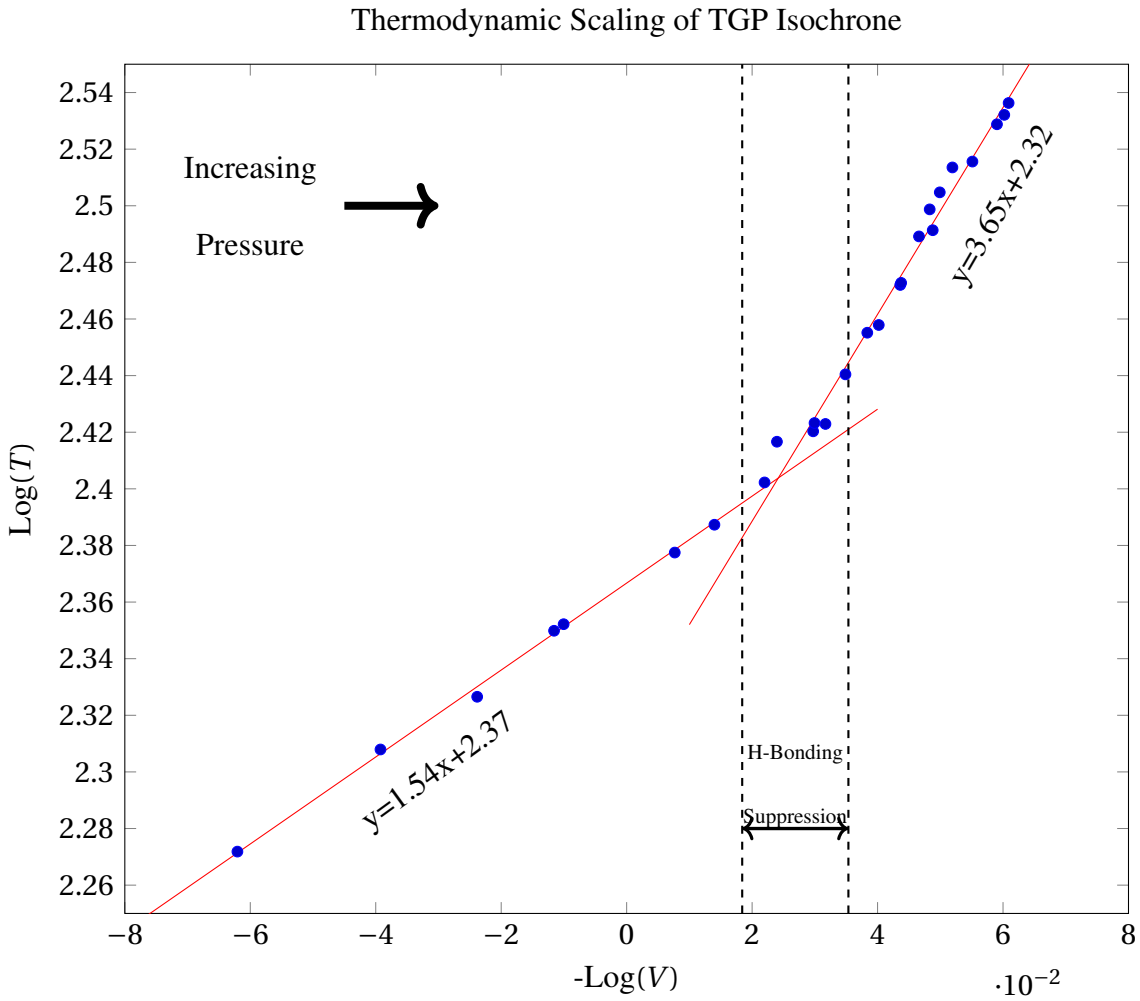


Figure 4.11: The linearized isochronal glycerol TGP data shows a transition from an H-bonded liquid at low pressure with a corresponding scaling factor of $\gamma \approxeq 1.54$ to a liquid which behaves more like one experiencing a simple Van der Waals potential at higher pressures, where H-boding is suppressed, exhibiting a scaling factor of $\gamma \approxeq 3.65$. The region between the dotted vertical lines corresponds the region indicated in Fig. 4.4 previously. The intersection of the two linear regimes is self consistent with the results of Pronin at approximately 2.5 GPa. It is also worthy of note that the average y-intercept of 2.35 implies a TGP isochrone of $\tau \approx 220$ s

In Fig. 4.5, the glycerol TGP data provided in App. B is linearized and plotted to determine the scaling parameter. The region between the dotted lines once again corresponds to that in Fig. 4.4—where pressure suppression of hydrogen bonding becomes prominent. The results are quite compelling. Two distinct regions emerge. At low pressure we find a scaling factor $\gamma = 1.54$ consistent with literature values. At high pressures, well within the region where hydrogen bonding

is assumed to be sufficiently suppressed, we find a significantly greater slope corresponding to $\gamma = 3.65$ consistent with what might be expected of a simpler Van der Waals liquid. A clear transition occurs between 2 and 3 GPa—consistent with our fragility results from our Andersson-Andersson contained surface model as well as Pronin's analysis.

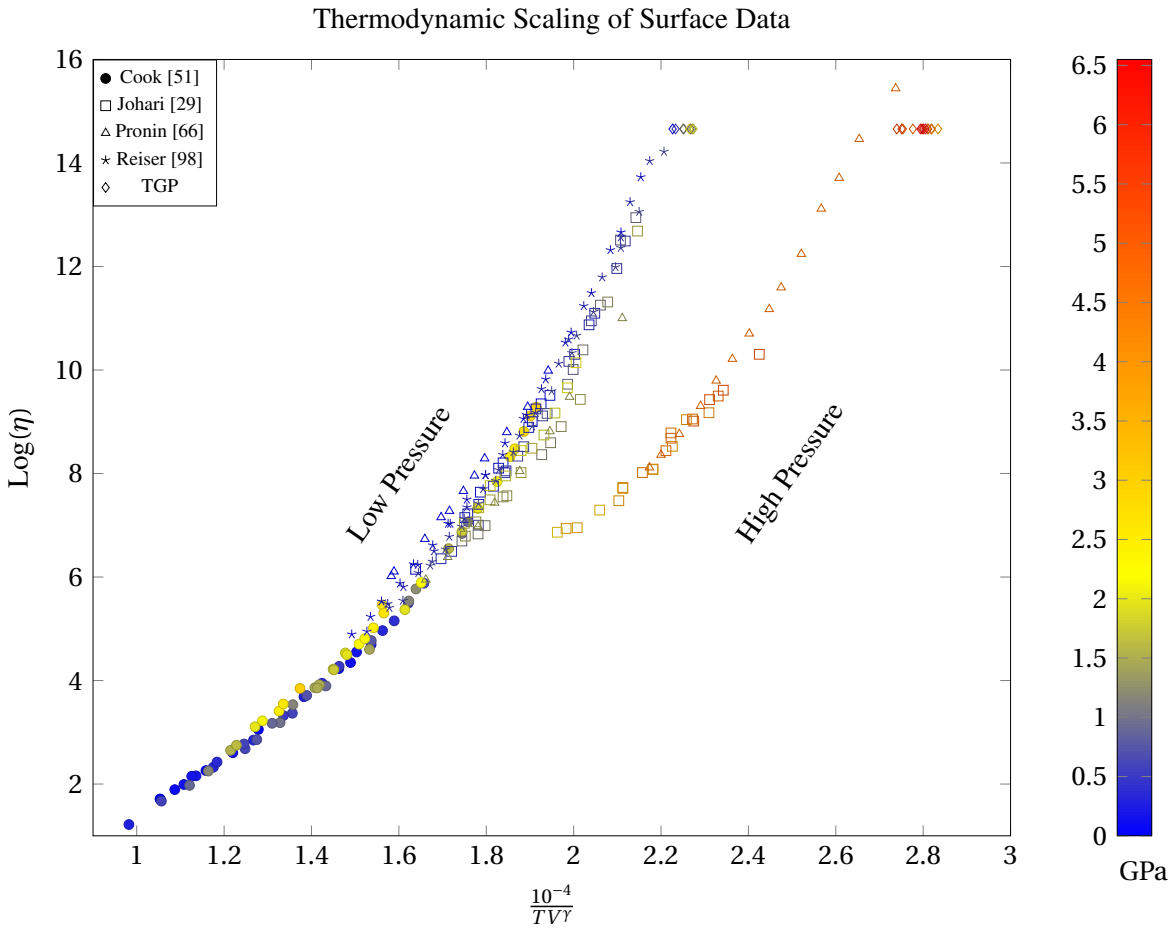


Figure 4.12: The results of thermodynamic scaling of glycerol for all low pressure data (< 2 GPa) with a scaling factor of $\gamma = 1.54$ and high pressure data (> 3 GPa) with a scaling factor of $\gamma = 3.65$.

All of the glycerol data that falls outside of the transition region between 2 and 3 GPa provided by Cook, Johari, Pronin, Reiser, and our TGP data exhibit a reasonable degree of thermodynamic scaling along one of two curves. The results of this scaling are provided in Fig. 4.5. We see that all of the Cook data scales nicely even well into the intermediate pressure regimen-perhaps

due to the relative high temperatures allowing for molecular mobility which facilitates hydrogen bonding at these higher pressures. The Johari data, on the other hand, is all at relatively low to intermediate temperatures and a clear pressure dependent divergence from good thermodynamic scaling is observed as pressure increases and hydrogen bonding begins to be suppressed. All of this still relies on an extreme extrapolation of the Tait equation of state for which Bridgman only provides data up to approximately 1.2 GPa however thermodynamic scaling is possible, albeit in two distinct regimens each with different scaling factor, γ. Where hydrogen bonding is not being suppressed we find a scaling factor $\gamma = 1.54$ that reflects a liquid for which free-volume plays a diminished role in the dynamics as thermal activation is necessary to ionic bonds between neighbors. At very high pressures where hydrogen bonding is completely suppressed we find a scaling factor of $\gamma = 3.65$ that reflects a liquid governed much more strongly by free-volume and Van der Waals interactions.

Once gain we note that the high pressure Pronin data taken in the toroidal pressure cell seems to scale at lower pressures or higher temperatures however, as vitrification sets in at higher pressure the data peels away from the trend expected if thermodynamic scaling were to hold in precisely the same manner that it peeled away from the Andersson-Andersson constrained surface fit previously. It appears once again as if the reported pressures are lower than what the dynamics would imply. As this discrepancy is occurring only at the highest pressure and in the region where vitrification is rapidly increasing, it stands to reason that they have underestimated the prominence of pressure gradients within their toroidal cell. The lower pressure data, even in the vicinity of the glass transition, scales nicely just as it conformed very well to the relaxation surface. However as pressures increase the odds of significant pressure gradients inside the cell increases greatly.